```
library(quantmod)
```

Financial Data Analysis in R Using “quantmod”

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DSS 445 - R Statistical Programming Language

Fall 2026

01 / About

What is quantmod?

- Created by **Jeffrey A. Ryan**; first released in August 2007
- Stands for **Quantitative Financial Modelling Framework**
- Built for traders and analysts to develop, test and chart financial models
- Used to quantitatively analyze financial time series data

Used for

- Downloading stock and economic data from Yahoo and FRED
- Creating professional-style Financial Charts
- Allows adding technical indicators (such as moving averages) to charts
- Calculating returns, dividends, volatility, etc.

02 / Data & Example

The Data

Yahoo Finance

Microsoft (MSFT)

Daily prices, 2007 to 2026

Open, High, Low, Close, Adjusted, Volume

FRED

U.S. Unemployment Rate (UNRATE)

Monthly rate, 2016 to 2026

Example

“How did Microsoft's stock hold up during COVID compared with the job market?”

Key Functions

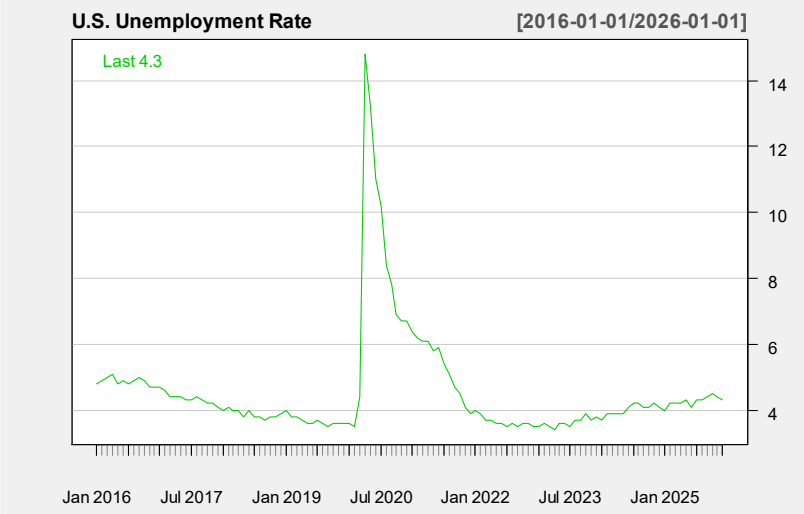
Function	What it does
getSymbols()	Downloads data from Yahoo Finance (default) or FRED
Cl() / Ad()	Extract the closing or adjusted price column (can also extract others)
chartSeries()	Draws line, bar or candlestick charts
addSMA()	Adds a moving average to the chart
yearlyReturn() / monthlyReturn()	Turns prices into percentage returns

04 / Results

What the Data Showed

- Microsoft stock returned 41% during 2020
- Unemployment peaked at **14.8%** in April 2020
- 2020 was one of Microsoft's best years in the stock market since its inception on the market.
- Meanwhile, unemployment shot up to the worst it has been in the past 10 years

```
> summary(Unemployment)
     Index                UNRATE
 Min.   :2016-01-01   Min.   : 3.400
 1st Qu.:2018-07-01   1st Qu.: 3.700
 Median :2021-01-01   Median : 4.100
 Mean   :2020-12-30   Mean   : 4.582
 3rd Qu.:2023-07-01   3rd Qu.: 4.700
 Max.   :2026-01-01   Max.   :14.800
                      NAs    :1
```




```
> # Step 5: Calculating Returns
> yearlyReturn(Microsoft)
           yearly.returns
2007-12-31     0.19023733
2008-12-31    -0.45393255
2009-12-31     0.56790117
2010-12-31    -0.08431758
2011-12-30    -0.06986746
2012-12-31     0.02889060
2013-12-31     0.40059907
2014-12-31     0.24164664
2015-12-31     0.19440255
2016-12-30     0.12004326
2017-12-29     0.37656907
2018-12-31     0.18739769
2019-12-31     0.55262378
2020-12-31     0.41039951
2021-12-31     0.51209428
2022-12-30    -0.28692911
2023-12-29     0.56800933
2024-12-31     0.12089137
2025-12-31     0.14737840
2026-10-09     0.10638520
```

Key Takeaways

- quantmod provides a simple way to import financial data from several sources for deep financial analysis
- Provides professional financial charts and indicators using a simple line of code
- Allows you to use basic R functions in order to easily analyze the data such as ways to extract data column by column.
- It is a very powerful tool for individuals interested in investing, trading or just looking for knowledge on the stock market.

Thank you!